

079 Make a weather front

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When a warm and a cool mass of air meet, they create a boundary, or “front” between them. You can’t see this happening in air, but you can make a front in a jar to see how it works.



What you need

- Glass jar • Pitcher
- Very hot (not boiling) water • Cold water
- Green food coloring
- Thermometer



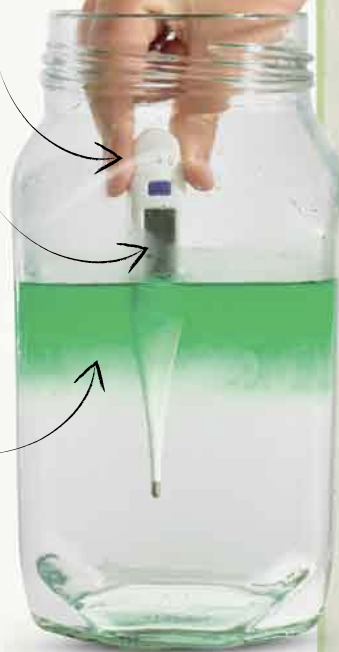
1 Fill the jar just over half-full with very cold water. Fill the pitcher with hot water. Add a few drops of food coloring. Stir.



2 Tip the jar of cold water so that the water is close to the rim. Now slowly pour the green hot water down the jar’s neck.

Hold the thermometer carefully

Test the temperatures of the layers



“Weather front”

3 The warm green water forms a layer above the cold, heavier water. The area where they meet acts like a weather front.

080 Make a rainbow

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Make a rainbow in your backyard. Stand with the Sun behind you. Set the hose on fine spray and slowly move it around until you see a rainbow appear.

Water droplets reflect and split sunlight into colors